Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 25.08.2026**

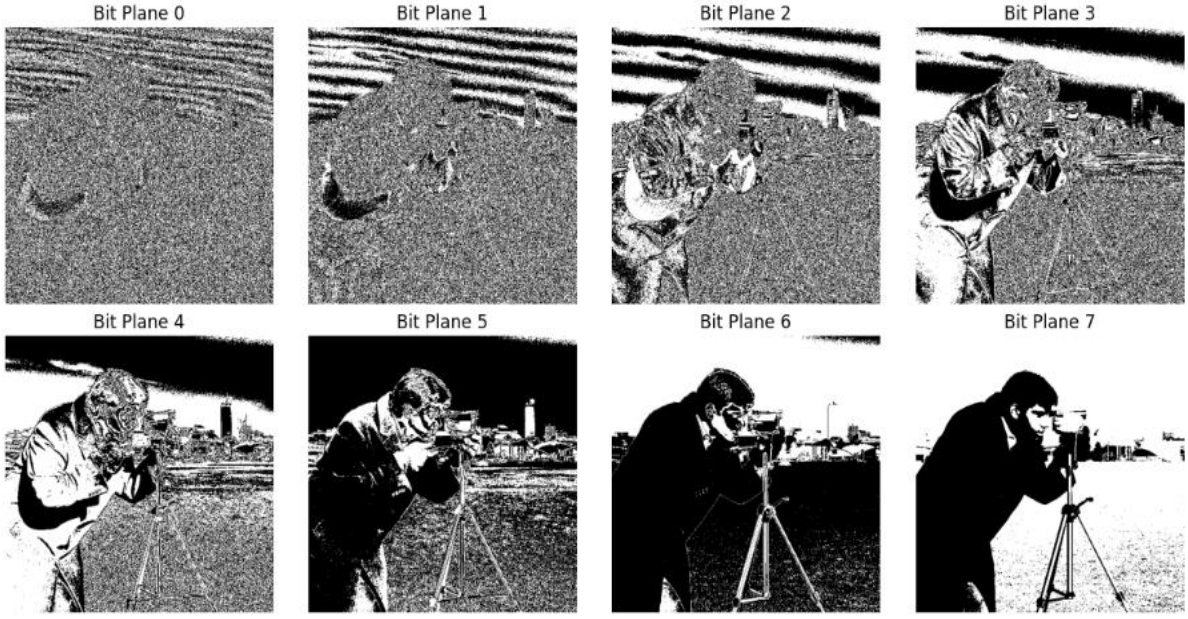
Assignment-3 (Image Enhancement-II)

1. Consider that the value of each pixel of a grayscale image can vary in the range $[0, 255]$. A fully black pixel is represented by '0', and a fully white pixel is represented by '255'. Any intermediate value represents different shades of grey. Since the pixel values lie between $0\ to\ 255$, information of each pixel is stored using $8 - bits$ binary number. So, we can decompose a grayscale image into 8 binary images by taking a single bit from a particular position of each pixel. Write a program that takes a grayscale image as input and produces all 8 binary images corresponding to each bit plane of the input image. **[10 Marks]**

Sample Input image:

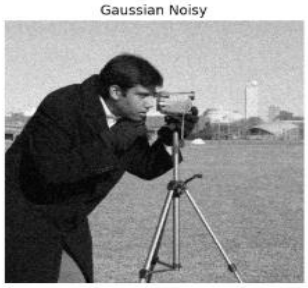


Output Image:



2. Suppose you have a grayscale image, "Gaussian_Noisy_Image.png", that is corrupted by Gaussian noise, and your task is to remove the noise to obtain a clean image. To remove the noise, write a function that takes the noisy image and a filter of size $k \times k$ as inputs and produces the denoised image as output. The filter should perform mean filtering, where each pixel is replaced by the mean value of the $k \times k$ neighborhood surrounding it. Apply the mean filter using three different filter sizes, $k = 3, 5, and\ 7$. Plot the corresponding denoised images obtained using each filter size and visually compare their effectiveness in reducing Gaussian noise. Finally, determine the optimal filter size based on the quality of the denoised output. *Do not use built-in libraries for filtering*. **[10 Marks]**

Sample Input:

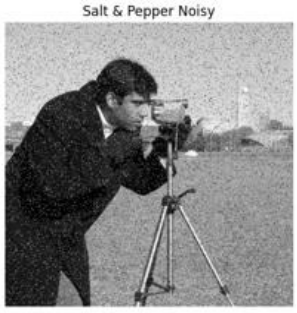


Sample Output:

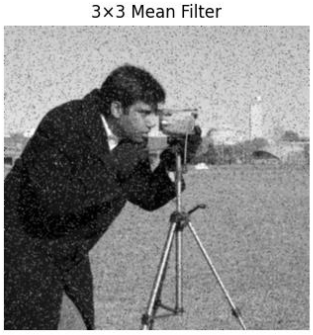


3. An image is corrupted with salt-and-pepper noise in "Salt_Pepper_Noisy_Image.png". The objective is to reduce the noise and obtain a clean image. You are required to implement two 3 × 3 spatial filters, namely, the mean filter and the median filter, using the pixel values within the corresponding 3 × 3 neighbourhood for noise removal. Compute the PSNR value of each filtered image with respect to the original image "cameraman.png". After computing the PSNR values, compare the performance of the mean and median filters. The filter that produces the higher PSNR value will be considered the best-performing filter for removing salt-and-pepper noise. For $Peak\ Signal - to - Noise\ Ratio$ (PSNR) $= 10\log_{10}(\frac{MAX_I^2}{MSE})$, where MAX_I is the maximum pixel intensity value, $Mean\ Squared\ Error\ (MSE) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}[I(x,y) - G(x,y)]^2$, and $M \times N$ is the dimension of the image, and $I\ (x, y)$ and $G\ (x, y)$ represent the original and filtered image pixel values, respectively. *Do not use built-in libraries for filtering and PSNR*. **[20 Marks]**

Sample Input:



Sample Output:





PSNR Values:

Mean Filter
MSE: 316.18
PSNR: 23.13 dB

Median Filter
MSE: 80.76
PSNR: 29.06 dB

Filter Comparison
Best Filter**:** Median Filter
PSNR**:** 29.06 dB